The Incompleteness of the Instantaneous State and the Status of Chapman-Kolmogorov Divisibility
Analysis from Limits, Relativity, and Renormalization*

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Abstract

Much of theoretical physics rests on the idealization that a physical system possesses a complete state at each instant of time—a state sufficient to determine all future evolution. We present independent arguments that this idealization is physically unjustified. First, the state at a temporal point is constructed as the limit of states over intervals; the ε-δ definition of a limit explicitly excludes the limit point from its own construction, guaranteeing convergence of the value in the neighborhood of the point while remaining formally silent on what properties the point itself carries. Second, the joint application of quantum mechanics and general relativity guarantees that the limiting process $\delta t \to 0$ physically self-destructs at the Planck scale, while quantum-mechanical clock bounds independently constrain the measurability of time intervals from below—two distinct mechanisms by which established physics prevents the temporal point from being physically realized. Third, quantum field theory has known since the 1940s that physics evaluated at a spacetime point is pathological—a fact encoded in ultraviolet divergences, systematized by the renormalization program, and given physical interpretation by Wilson's demonstration that the divergences reflect the finite domain of validity of any effective description, not a deficiency of any particular calculational method. Taken together, these arguments show that the completeness of the instantaneous state is not a consequence of how that state is defined but a separate physical assumption, and one that established physics gives positive reasons to reject. Any framework that presupposes it, including the Chapman-Kolmogorov factorization condition central to the theory of stochastic processes, inherits that liability. The burden therefore reverses: a memoryless, divisible description is the special case that must be justified, and indivisibility is the generic expectation.

Keywords: Chapman-Kolmogorov divisibility; indivisible stochastic processes; stochastic-quantum correspondence; instantaneous present; measurement problem; renormalization; Planck scale

1 Introduction

The assumption that a physical system possesses a complete state at each instant of time is so deeply embedded in theoretical physics that it is rarely stated as an assumption at all. In

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classical mechanics, the state is specified by positions and momenta at a time t; in quantum mechanics, by a vector in Hilbert space or equivalently a density matrix; in classical field theory, by field configurations and their conjugate momenta on a Cauchy surface. In every case, the formalism presupposes that the state at a temporal point carries the full kinematic specification of the system—all degrees of freedom, all constraints, all information required to determine the system's future evolution.

This paper argues that completeness is not a property the instantaneous state inherits from the way it is constructed, but a separate physical assumption that established physics contradicts. Three independent arguments support this. One, from the mathematics of limits, shows that the construction defining the instantaneous state does not deliver the full kinematic specification a dynamical law requires. The second, from relativistic gravitation and quantum clock bounds, shows that the physical process of reaching the temporal point cannot be carried out. The third, from quantum field theory, shows that the point-valued limit is exactly where the established formalism becomes pathological. The claim is not that an instant is empty. It is that its completeness must be argued for rather than presupposed, and that the argument runs opposite to the usual direction: the divisible, memoryless description is the case requiring justification, while indivisibility is what one should expect by default.

This reversal has a precise target. The Chapman-Kolmogorov (CK) factorization condition requires the state at each intermediate time to act as a complete relay between past and future, carrying everything the subsequent evolution needs:

$$\Gamma(t_2 \leftarrow t_1) = \Gamma(t_2 \leftarrow t) \cdot \Gamma(t \leftarrow t_1), \tag{1}$$

for every intermediate time $t_1 < t < t_2$. When this factorization fails, the process is *indivisible*: the transition matrix over the whole interval is a primitive dynamical object, not decomposable through intermediate times. Barandes [1, 2, 3] has shown that these indivisible stochastic processes stand in exact correspondence with quantum mechanics, with interference, entanglement, decoherence, and the Born rule arising from indivisibility alone. That program establishes the correspondence as a formal result and largely takes indivisibility as a premise. The contribution here is to supply its physical motivation. If the instantaneous state does not carry the information CK factorization assumes, indivisibility is not an exotic condition to be imposed but the default, and the Markovian factorization is the idealization in need of defense. Each of the three arguments below is drawn from established physics; what is new is that they are made to bear jointly on the information content of the temporal point, and through it on the status of CK divisibility.

The question is not whether the state at a temporal point can be *defined*. It can, and nothing here disputes the definition. The question is whether the object so defined inherits the property dynamical frameworks demand of it, the complete kinematic information sufficient to fix future evolution. The construction does not supply that property, and established physics provides independent reasons to deny it.

2 Argument from Limits

The state of a physical system at a temporal point is not directly observable [8]: every physical measurement ever performed has yielded an average over a nonzero interval. The "state at time t" is a mathematical extraction: the limit of interval-averaged states as the interval shrinks to zero. In standard analysis, the infinitesimal interval has no existence; one has either finite intervals or the point, with nothing in between. The question is therefore what the point inherits from the intervals that define it.

The ε-δ definition of this limit is mathematically rigorous [4]. The limit value exists and is well-defined. This is not in dispute.

What is in dispute is whether the limit point inherits the properties that dynamical frameworks demand of it. Consider what the ε-δ definition actually guarantees. For a function f and a point a, the statement $\lim_{x\to a} f(x) = L$ means: for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all x with $0 < |x - a| < \delta$, we have $|f(x) - L| < \varepsilon$. The condition $0 < |x - a|$ is essential: the definition explicitly excludes the point a itself. The limit characterizes the behavior of f in every punctured neighborhood of a while making no assertion about a itself.

This exclusion is not a technicality. It is the reason limits are useful: they allow us to assign a value at a point where the function may not be defined, or where its value may differ from the limiting behavior. The limit of $\sin(x)/x$ as $x \to 0$ is 1, but the function is undefined at $x = 0$. The limit construction delivers the value; it does not deliver the function's other properties at the point, because the point is excluded from the construction.

That the $\delta t \to 0$ limit can fail to deliver physically expected quantities is not a new observation. As Landau and Lifshitz emphasized [7], in quantum mechanics the ratio $\Delta x/\Delta t$ does not converge as $\Delta t \to 0$: successive position measurements become more erratic, not less, and velocity—defined as the limit of this ratio—does not exist. The limiting process fails to deliver even a derived kinematic quantity. The present argument extends this logic to the state itself: if the limit already fails for the derivative, the assumption that it delivers the full kinematic specification at the point requires justification.

The physical consequence is direct. The state at a temporal point is constructed as a limit of interval-averaged states. The ε-δ definition guarantees that these interval-averaged states converge to a well-defined value. But the full kinematic specification of a physical system—the complete set of degrees of freedom, their interrelations, and the constraints that govern their evolution—is a richer structure than a single value. The definition of a limit is silent on whether this richer structure survives at the limit point, because the limit point is excluded from the construction that defines it.

This argument applies within the standard continuum topology of the real line. Alternative topological structures—non-Hausdorff spacetimes in which limits need not be unique [5], or causal set theory in which the continuum is replaced by a discrete partial order [6]—would modify the analysis in ways specific to each framework. The present argument targets the standard idealization.

This does not prove that the limit point lacks the required kinematic information. It establishes that nothing in the mathematical construction guarantees it has that information. The completeness of the state at a temporal point is therefore not a consequence of how such states are constructed. It is an additional assumption—one that, as the following sections show, contradicts established physics.

3 Arguments from Relativity

General relativity and quantum mechanics are among the most precisely confirmed theories in the history of science. Applied jointly, they yield a concrete prediction about the limits of spatial measurement through two independent lines of argument.

Mead [9] and Garay [10] establish that a fundamental minimal spatial length must exist as a consequence of the joint requirements of quantum mechanics and general relativity. The argument is forced by the uncertainty relations: through the de Broglie relation, improving spatial resolution to a scale δx requires concentrating momentum of order $\hbar/\delta x$, and therefore

energy of order $\hbar c/\delta x$, within the probed region. At sufficiently small δx the Schwarzschild radius of this energy exceeds δx itself, and the region undergoes gravitational collapse. Since the fundamental length deduced by Mead scales as $\sqrt{G}$, the Planck length $l_P = \sqrt{\hbar G/c^3} \approx 1.6 \times 10^{-33}$ cm—the unique length constructible from $\hbar$, G, c—is the natural lower bound, and both analyses yield results consistent with this scale, though neither derives it as a sharp numerical threshold. The existence of this fundamental length in turn limits the precision with which clocks can be synchronized, as Mead shows [9]—an observation originally motivated by the problem of infinities in quantum field theory arising from point-particle idealizations, connecting the gravitational argument directly to the renormalization evidence of Sec. 4.

't Hooft's analysis of trans-Planckian scattering [11] arrives at the same spatial obstruction by a route that is arguably more direct: when two particles collide at sufficiently high center-of-mass energy, general relativity requires that the interaction region undergo gravitational collapse. This conclusion depends only on the particle kinematics and the Einstein equations, without recourse to quantum interference effects, and therefore rests on fewer assumptions than the Mead-Garay derivation.

A third, independent route to a temporal limitation is provided by Salecker and Wigner [12], who show that quantum mechanics imposes fundamental bounds on the precision of any clock used to measure spacetime intervals. An ideal clock measuring a time interval T with resolution δt must, to avoid quantum spreading of its center of mass degrading its accuracy, satisfy a minimum mass condition $M \gtrsim \hbar T/(c^2\,\delta t^2)$. Combining this with the requirement that the clock not collapse into a black hole—$r_s = 2GM/c^2 \lesssim \ell_{\text{clock}}$—yields $\delta t \gtrsim t_P$. The argument proceeds entirely from the quantum-mechanical properties of the clock itself, without invoking the gravitational self-collapse of the measured region, and therefore rests on physical grounds distinct from those of either Mead or 't Hooft.

The temporal consequence of the spatial results follows independently. Since the Planck time

$$t_P = \frac{l_P}{c} = \sqrt{\frac{\hbar G}{c^5}} \approx 5.4 \times 10^{-44}\ \text{s} \tag{2}$$

is the light-crossing time of the Planck length, probing temporal intervals shorter than t_P requires concentrating energy at the same densities that trigger gravitational self-collapse in the Mead and 't Hooft analyses. This temporal extension is not stated by Mead, Garay, or 't Hooft, but is a direct physical inference from their spatial results: the energy scales are identical, related by the factor c. The Mead and 't Hooft analyses, together with this inference, imply that the interaction region collapses into a black hole before the temporal measurement completes. The Salecker-Wigner argument arrives at the same temporal bound from an independent direction: the quantum-mechanical properties of the clock itself prevent arbitrarily fine temporal resolution, without any appeal to the gravitational collapse of the measured region.

We stress that this result follows from fundamental considerations rather than modelling assumptions: it is a consequence of applying two established theories simultaneously, and does not rely on the time-energy uncertainty relation, whose interpretation is complicated by the fact that time is not an observable in standard quantum mechanics. The specific physics below t_P is unknown. What is known is that the limiting process $\delta t \to 0$ cannot be physically completed. The temporal point is a limit that established physics prevents from being physically realized.

Approaches to quantum gravity that begin with a discrete structure at the Planck scale—lattice gravity, causal set theory, spin foams—are compatible with this conclusion: they abandon the continuum temporal point from the outset, replacing the standard idealization with a fundamentally different mathematical structure in which the instantaneous present, as conventionally conceived, plays no role.

The preceding section established that the mathematical construction of the point-state

does not guarantee dynamical completeness at the point. This section establishes that the physical processes required to reach the point cannot be carried out: gravitational self-collapse and quantum clock bounds independently prevent the limiting process from being physically completed.

4 Argument from Renormalization

Quantum field theory provides corroboration from an independent direction. When interactions in QFT are evaluated at a spacetime point—taking the separation between interacting fields to zero—the result is infinite. These ultraviolet divergences are well understood: they arise whenever the formalism is pushed to the point-valued limit [13].

The renormalization program [14, 15, 16, 17, 18, 19], given conceptual foundation by Wilson [17, 18, 19], extracts finite physical predictions by introducing a regulator that prevents the theory from reaching the spacetime point. The physical content is obtained at finite resolution and shown to be independent of the specific regulator. But the point-valued limit is never taken. The resulting predictions agree with experiment to extraordinary precision [22]—not because the point-valued limit is well-behaved, but because the physics at finite resolution is sufficient and self-consistent without it.

It is essential to distinguish the Wilsonian understanding of renormalization from the earlier view in which divergences were regarded as computational artifacts of perturbative methods. Wilson's framework demonstrates that the need for renormalization is not an artifact of perturbation theory. Lattice quantum field theory, which is a fully non-perturbative formulation, requires renormalization for the same structural reasons [20]. The running of coupling constants with energy scale—for instance, the running of the strong coupling α_s in quantum chromodynamics—is an experimentally measured phenomenon [21], not a calculational convenience. In Wilson's picture, a theory at energy scale μ is an effective description whose parameters run with the scale, obtained by integrating out degrees of freedom above that scale. The "bare" theory at a spacetime point is not a physically meaningful object that we have failed to calculate correctly, but the boundary where the effective description ceases to be valid. This is why even the lattice formulation—which replaces the continuum with a finite number of degrees of freedom—still requires renormalization: physical quantities depend on how those degrees of freedom are coarse-grained, regardless of whether the underlying structure is continuous or discrete. The renormalization group is the systematic expression of this fact.

Physics at a spacetime point is where the formalism breaks down. Physical content is extracted at finite resolution. Whether a future theory could render the spacetime point well-defined is an open question; but the argument here is evidential, not deductive. Our best established theories—the very theories whose predictions have been confirmed to extraordinary precision—provide no basis for assuming that the point carries complete physical information, and positive grounds for doubting it. This conclusion finds independent support in algebraic quantum field theory, where the Haag-Kastler axioms assign observable algebras to open spacetime regions and the algebra at a point is provably trivial [24]; a rigorous, non-perturbative result that merits separate treatment. The parallel with the preceding arguments is not analogical but structural: in all three cases, the temporal or spacetime point is where the physics fails, and the meaningful content lives at finite resolution.

5 Dissolution of the Measurement Problem

The measurement problem is standardly formulated as the tension between the deterministic, unitary evolution of the quantum state and the occurrence of definite, stochastic outcomes upon measurement [26]. This tension presupposes that both sides of the measurement event—the unitarily evolving superposition before and the definite outcome after—are complete states at temporal points, where "complete" means carrying sufficient dynamical information to determine future evolution (whether pure or mixed). The demand is that the timeline can be cut at "the moment of collapse" with a fully specified state on either side. This is precisely the demand that the instantaneous present carry complete dynamical information, applied to the measurement event.

Measurement is a physical interaction between system and apparatus. Every stage—photons hitting detectors, electrons cascading through amplifiers, pointers moving on dials—takes nonzero time. There is no instant at which the measurement happens. There is an interval over which system and apparatus become correlated—a picture independently supported by decoherence theory, which models the finite timescale over which system-environment entanglement suppresses interference [25, 26]. At the beginning, they are uncorrelated. At the end, the apparatus registers a definite outcome.

Asking "at which point in between did the transition occur?" is demanding dynamical completeness at intermediate temporal points within the measurement interval. The preceding arguments show this demand is unjustified: the mathematical construction does not guarantee completeness at the point, the physical limit to the point cannot be reached, and the point-valued limit is where known physics becomes pathological.

The collapse-timing formulation of the measurement problem is thereby dissolved rather than solved. The apparent conflict between unitary evolution and definite outcomes depends on both being complete descriptions at temporal points; if temporal points do not carry complete dynamical information, the conflict has no well-defined terms in which to arise. Other aspects of the measurement problem—the derivation of the Born rule, the definite-outcome problem, and the preferred-basis problem—are not addressed by this dissolution and require independent treatment.

The specific outcome of a measurement—why *this* result rather than *that* one—is not addressed by this dissolution, nor does it need to be. Within the framework of indivisible stochastic processes [1, 2, 3], the outcome is a stochastic draw from the transition matrix over the measurement interval. It is irreducibly stochastic. In the standard Hilbert space formulation, the Schrödinger equation is deterministic, making stochastic outcomes appear to require a mechanism—collapse—that violates the governing equation. In the indivisible stochastic formulation, the dynamics are stochastic from the start; the transition matrix yields probabilities and outcomes are draws from those probabilities, with no deterministic equation being violated. As Barandes establishes [1, 2], the Schrödinger equation deterministically governs the evolution of the transition matrix—the probability law itself—not the realization of individual outcomes, which was never deterministic in either formulation. The two formulations are empirically equivalent—they produce identical predictions. But the conceptual puzzle is an artifact of the Hilbert space formulation's mathematical structure, not of the physics.

6 Consequences

If the instantaneous present does not carry complete dynamical information, then any factorization condition that presupposes it—including the Chapman-Kolmogorov condition, Eq. (1)—is

an idealization rather than a physical fact. Through Barandes's stochastic-quantum correspondence [1, 2, 3], this means indivisibility is the generic dynamical condition and quantum behavior is the default, not the exception.

Classical behavior emerges when the observer's temporal integration window δt_{obs} satisfies $\delta t_{\text{obs}} \gg \tau_{\text{coh}}$, where τ_{coh} is the system's coherence time—the timescale over which indivisibility is manifest. In this regime, violations of the factorization condition are integrated out and the process appears divisible. The quantum-to-classical transition is a consequence of finite temporal resolution, not an ontological boundary.

This yields a falsifiable prediction: any stochastic system observed at sufficiently fine temporal resolution relative to its coherence time should exhibit measurable violations of the CK condition. Recent work [23] reports CK violations in financial time series, consistent with this prediction, though the authors interpret the violations as memory effects rather than indivisibility. Systematic measurements across multiple physical systems—mesoscopic devices, turbulent flows, biological processes—would constitute a direct test.

7 Discussion

The three arguments assembled here draw only on established mathematics and physics, and none invokes a particular interpretation of quantum mechanics, proposes new axioms, or modifies existing theory. The argument from limits requires only the ε-δ definition of a mathematical limit. The argument from relativity requires the established validity of quantum mechanics and general relativity, where gravitational self-collapse and Salecker-Wigner clock bounds independently prevent the temporal point from being physically reached. The argument from renormalization requires only the existence of ultraviolet divergences and their physical interpretation through Wilson's renormalization group. What is new is not any of these results but the use to which they are jointly put: together they remove the basis for treating the completeness of the instantaneous state as given, and they do so without leaving the domain of confirmed physics.

The consequence is a reversal of the burden of proof. The Chapman-Kolmogorov factorization condition requires the state at each intermediate time to relay the whole of the dynamics, and that requirement is exactly the completeness the preceding arguments deny. Once completeness is no longer presupposed, divisibility is the special case that must be justified and indivisibility is the generic expectation, which through Barandes's correspondence is to say that quantum behavior is the default and classical, divisible dynamics the approximation recovered at coarse temporal resolution. The dissolution of the collapse-timing measurement problem follows as one instance of this reversal rather than as a separate result. The claim the paper leaves standing is therefore not about the metaphysics of the instant but about the status of a factorization condition: Chapman-Kolmogorov divisibility is an idealization in need of defense, not a structural feature of stochastic dynamics that can be assumed.

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